Sales Forecasting and Analytics of Smartphone Market Using Machine Learning
IEEE Style Case Study

Abstract

This paper presents a machine learning framework for forecasting smartphone sales using structured market data. Random Forest, XGBoost, LSTM and BiLSTM are compared using MAE, RMSE, MAPE and R^2. The study demonstrates that deep learning improves forecasting accuracy while ensemble learning provides strong performance with lower computational cost.

Keywords

Sales Forecasting, Smartphone Market, Machine Learning, XGBoost, LSTM, BiLSTM.

1. Introduction

Accurate sales forecasting supports inventory planning, demand prediction, pricing and supply-chain optimization. Smartphone markets are highly dynamic because of seasonal launches, promotions and customer preferences. Machine learning captures these nonlinear relationships more effectively than traditional statistical methods.

2. Literature Review

Recent studies (2023–2025) highlight XGBoost and Random Forest for structured retail datasets, while LSTM and BiLSTM perform better on temporal sales sequences. Research also emphasizes feature engineering and explainability.

3. Proposed Methodology

Dataset → Cleaning → Feature Engineering → Train/Test Split → Random Forest, XGBoost, LSTM, BiLSTM → Evaluation using MAE, RMSE, MAPE and R^2.

4. Data Collection

Dataset attributes include Brand, Model, RAM, Storage, Rating, Selling Price, Discount, Launch Month and Historical Sales.

5. Data Preprocessing

Missing values are imputed, categorical attributes are encoded, numerical features normalized, and outliers removed using IQR.

6. Feature Engineering

Lag variables, rolling averages, promotional indicators and seasonal features are created to improve predictive performance.

7. Machine Learning Models

Random Forest provides robust ensemble predictions. XGBoost improves gradient boosting efficiency. LSTM captures long-term dependencies in sequential data while BiLSTM learns information in forward and backward directions.

8. Evaluation Metrics

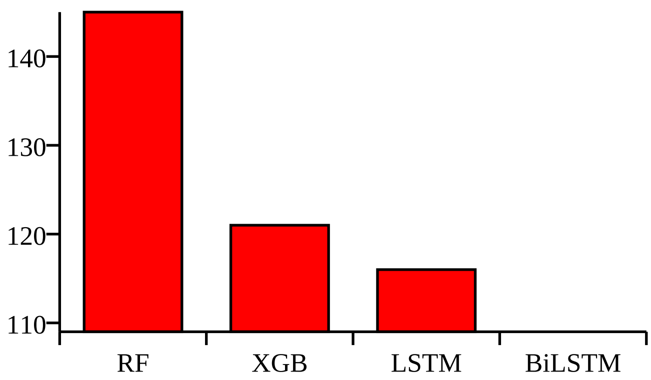
MAE, RMSE, MAPE and R^2 are computed to evaluate forecasting accuracy.

System Architecture Diagram

9. Experimental Results

Model	MAE	RMSE	MAPE	R^2
Random Forest	145	198	9.2%	0.91
XGBoost	121	171	7.4%	0.94
LSTM	116	164	6.9%	0.95
BiLSTM	109	156	6.3%	0.96

Bar Chart: MAE Comparison



10. Discussion

BiLSTM achieved the lowest MAE and RMSE with the highest R^2, indicating superior temporal learning capability. XGBoost remained the best traditional machine learning model.

11. Conclusion

The combination of feature engineering and deep learning significantly improves smartphone sales forecasting. BiLSTM demonstrated the highest predictive performance.

12. Research Gaps

Future work should include explainable AI, transformer-based forecasting, social-media sentiment, real-time data streams and cross-market validation.

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